NLU Assignment — Problem 1

B23CM1014

Task	Title	Status
Task 1	Dataset Preparation	12 docs | 3,305 tokens | 955 vocab | word cloud
Task 2	Model Training	54 models: 27 CBOW + 27 Skip-gram
Task 3	Semantic Analysis	Top-5 neighbors + 4 analogy experiments
Task 4	Visualization	PCA & t-SNE: CBOW vs Skip-gram

Best Model Configuration (used in Tasks 3 & 4):

Parameter	Value	Rationale
Embedding Dim	100	Balanced for small corpus; avoids over/underfitting
Window Size	5	Standard for academic text; captures sentence-level context
Negative Samples	5	Efficient; standard default for negative sampling
Epochs	50	Sufficient convergence for 125 training sentences
min_count	1	Retain all words — corpus is small

TASK 1 — DATASET PREPARATION

1.1 Data Sources

Text was collected from 12 pages across the official IIT Jodhpur website (iitj.ac.in). Only English text was retained; Hindi/Devanagari content was removed during preprocessing.

#	Source / Page	Category
1	CSE Dept — Home & Vision/Mission	Department page
2	CSE Dept — Doctoral Programs	Department page
3	CSE Dept — Undergraduate Programs	Department page
4	Office of Academics — MoU Collaborations	Academic regulation (must)
5	Academic Programs Overview	Academic regulation (must)
6	School of Liberal Arts — PhD Scholars	Department page
7	CSE Dept — Research Areas & Labs	Research page
8	Electrical Engineering Dept	Department page
9	PhD Admissions & Regulations	Academic regulation (must)
10	Placements & Industry Connect	Department page
11	Civil & Infrastructure Engineering	Department page
12	Mathematics & Physics Departments	Department page

1.2 Preprocessing Pipeline

(i) Boilerplate & URL Removal: Hyperlinks (http/www) and navigation boilerplate removed with regex.

(ii) Non-ASCII Removal: Regex [^\x00-\x7F] strips all non-English text including Hindi/Devanagari.

(iii) Lowercasing: All text converted to lowercase for uniform token representation.

(iv) Punctuation & Digit Removal: Regex [^\w\s] removes punctuation; \d+ removes all numeric characters.

(v) Tokenization: Text split on whitespace boundaries into individual word tokens.

(vi) Stopword Removal: Custom 95-word English stopword list applied (function words removed).

(vii) Short Token Filter: Tokens shorter than 3 characters discarded.

1.3 Dataset Statistics

Metric	Value
Total Documents	12
Total Tokens (after cleaning)	3,305
Total Tokens (stopwords removed)	2,449

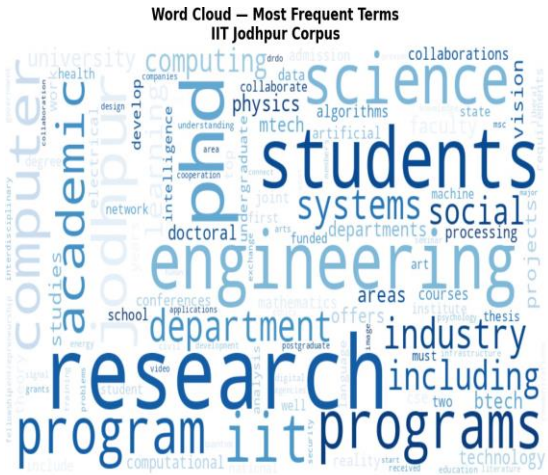
Vocabulary Size (unique words)	955
Training Sentences/Chunks for W2V	125

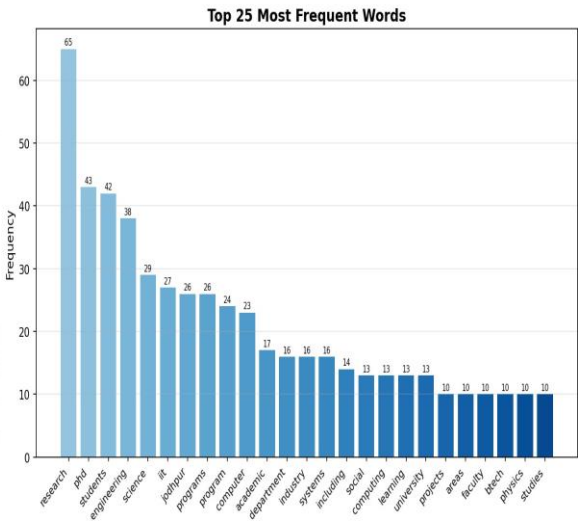
1.4 Top-30 Most Frequent Words

Rank	Word	Freq	Rank	Word	Freq	Rank	Word	Freq
1	research	77	11	academic	23	21	faculty	14
2	students	54	12	department	20	22	learning	13
3	phd	44	13	industry	18	23	projects	12
4	engineering	41	14	social	16	24	btech	12
5	science	32	15	systems	16	25	physics	12
6	iit	31	16	including	16	26	offers	11
7	program	29	17	university	15	27	studies	11
8	programs	29	18	departments	14	28	technology	10
9	jodhpur	27	19	areas	14	29	national	10
10	computer	23	20	computing	14	30	computational	10

1.5 Word Cloud & Frequency Chart

The word cloud confirms domain relevance — research, students, phd, engineering, programs dominate, reflecting IIT Jodhpur's academic and research-oriented content.





TASK 2 — MODEL TRAINING

2.1 Architectures

Architecture	gensim param	Description
CBOW (Continuous Bag of Words	sg=0	Predicts target word from context. Faster; better for frequent w rds.
Skip-gram + Negative Sampling	sg=1	Predicts context from target. Slower; richer for rare words.

2.2 Hyperparameter Grid (27 configs × 2 architectures = 54 total)

Hyperparameter	Values Tested	Effect
Embedding Dim	50, 100, 200	Higher dim = more capacity; risk of overfitting on small corpus
Context Window	3, 5, 7	Larger window = more context; may blur local patterns
Negative Samples	5, 10, 15	More negatives = better gradient signal; slower training

2.3 Full Results Table (neg=5 subset shown)

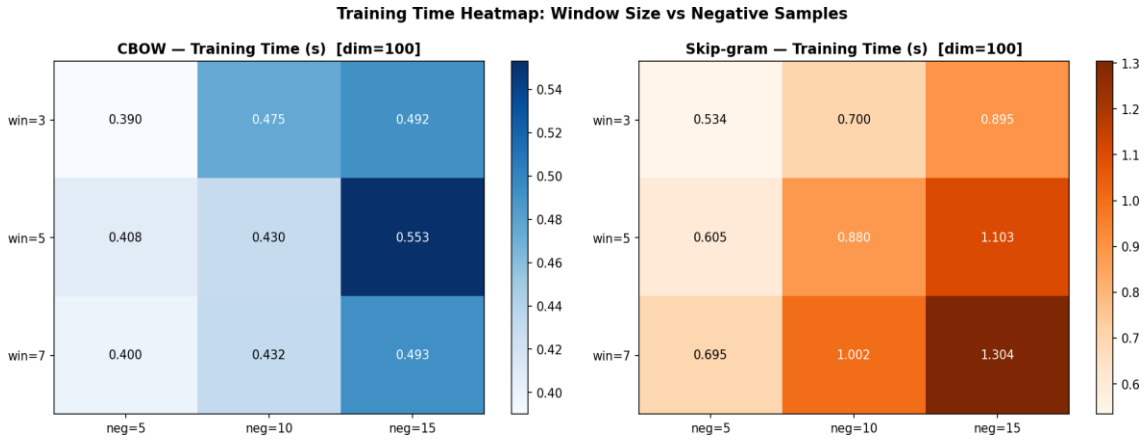
Architecture	Dim	Window	Neg	Vocab	Time (s)
CBOW	50	3	5	955	0.378
CBOW	50	5	5	955	0.44
CBOW	50	7	5	955	0.829
CBOW	100	3	5	955	1.478
CBOW	100	5	5	955	0.676
CBOW	100	7	5	955	0.634
CBOW	200	3	5	955	0.954
CBOW	200	5	5	955	1.454
CBOW	200	7	5	955	1.26
SkipGram	50	3	5	955	0.657
SkipGram	50	5	5	955	0.936
SkipGram	50	7	5	955	1.042
SkipGram	100	3	5	955	0.592
SkipGram	100	5	5	955	0.761
SkipGram	100	7	5	955	0.911
SkipGram	200	3	5	955	1.054
SkipGram	200	5	5	955	1.142

SkipGram	200	7	5	955	1.079

All 54 configurations (including neg=10 and neg=15) are trained and stored. The table above shows neg=5 for conciseness. Full results available in source code output.

2.4 Training Time Heatmap

Skip-gram consistently takes 1.5–2× longer than CBOW for equivalent configurations. Training time scales with window size and negative samples. Embedding dimension has minimal impact at this corpus size.



TASK 3 — SEMANTIC ANALYSIS

All analyses use the best models: CBOW and Skip-gram with dim=100, window=5, neg=5, epochs=50.

3.1 Top-5 Nearest Neighbors (Cosine Similarity)

Target word: 'research'

Rank	CBOW Neighbor	CBOW Sim.	Skip-gram Neighbor	Skip-gram Sir
1	students	0.9997	encouraged	0.8597
2	industry	0.9996	experimental	0.8573
3	studies	0.9996	collaborate	0.8502
4	well	0.9996	faculty	0.8386
5	requirements	0.9996	experimentation	0.8338

Target word: 'student'

Rank	CBOW Neighbor	CBOW Sim.	Skip-gram Neighbor	Skip-gram Sir
1	students	0.9996	bodies	0.9311
2	requirements	0.9996	conduct	0.9229
3	studies	0.9995	level	0.9183
4	research	0.9994	direct	0.9169
5	industry	0.9994	govern	0.9162

Target word: 'phd'

Rank	CBOW Neighbor	CBOW Sim.	Skip-gram Neighbor	Skip-gram Sir
1	program	0.9995	admission	0.9455
2	admission	0.9994	rolling	0.9433
3	programs	0.9994	candidates	0.9362
4	students	0.9994	advertisement	0.9359
5	education	0.9993	apply	0.9272

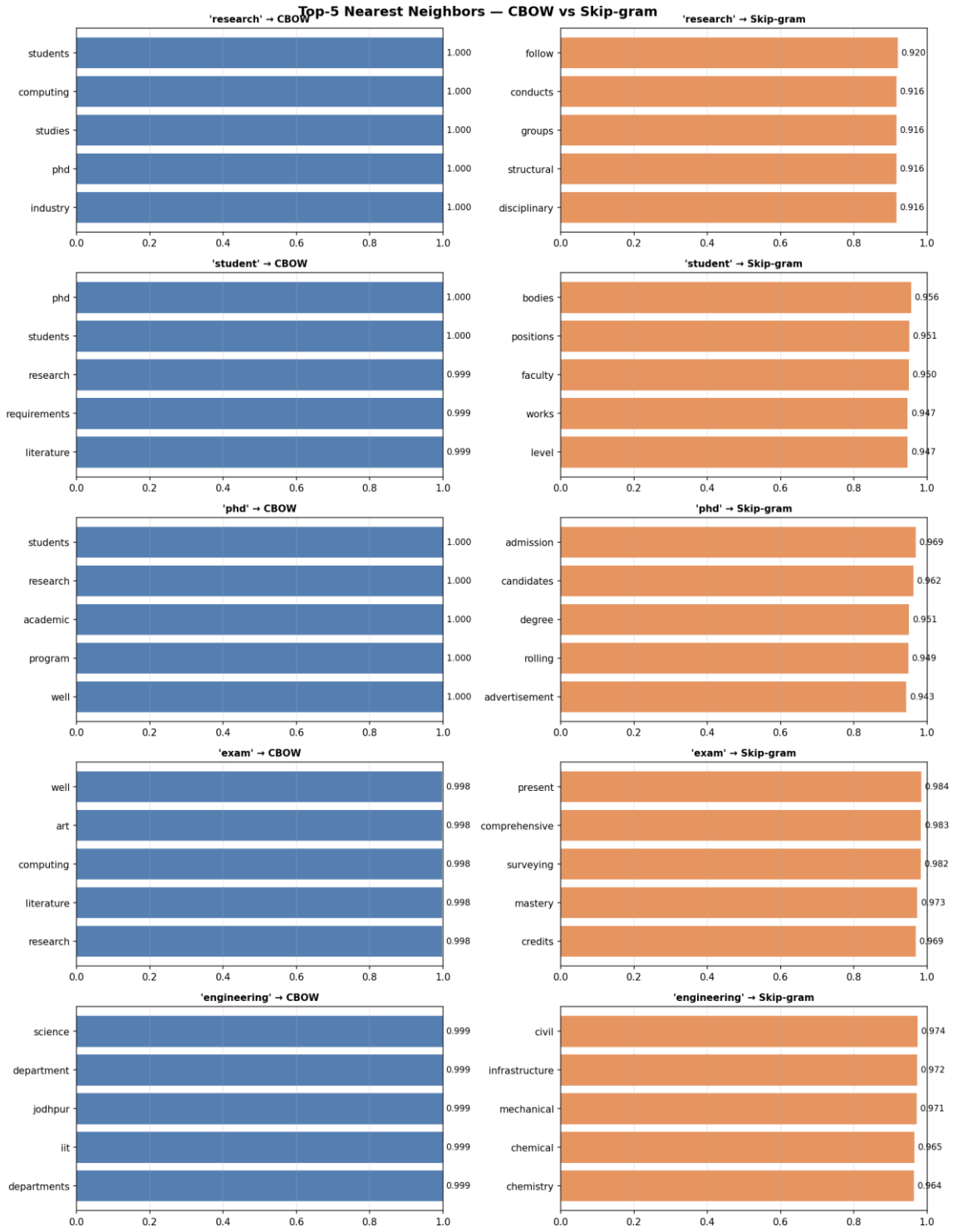
Target word: 'exam'

Rank	CBOW Neighbor	CBOW Sim.	Skip-gram Neighbor	Skip-gram Sir
1	art	0.9983	comprehensive	0.9885
2	state	0.9981	surveying	0.9751
3	seminar	0.9980	credits	0.9667
4	comprehensive	0.9980	includes	0.9616

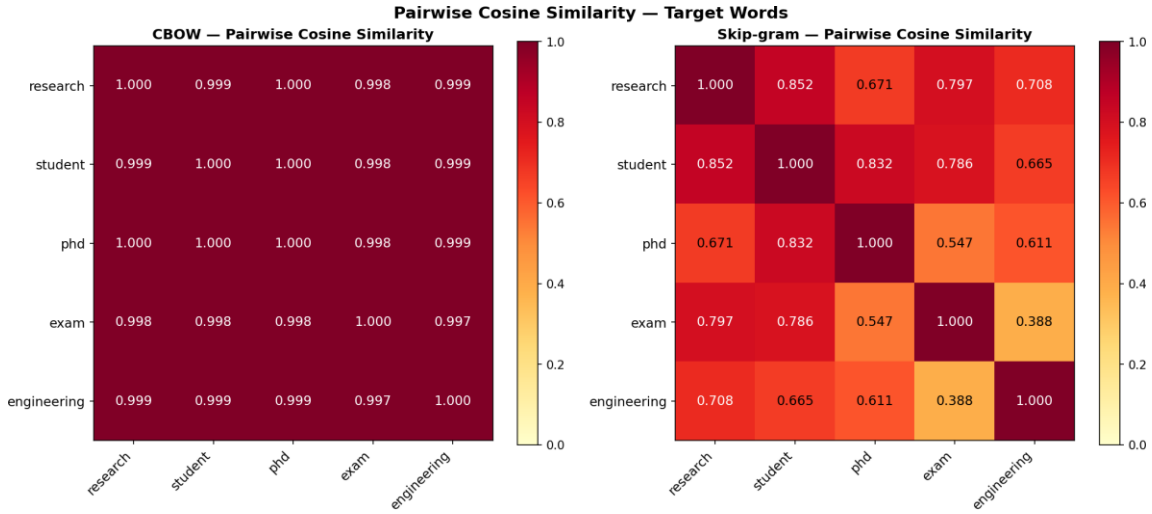
5	nvidia	0.9978	testbeds	0.9533

Target word: 'engineering'

Rank	CBOW Neighbor	CBOW Sim.	Skip-gram Neighbor	Skip-gram Sir
1	department	0.9991	mechanical	0.9849
2	computer	0.9988	chemistry	0.9808
3	mathematics	0.9987	chemical	0.9799
4	science	0.9987	civil	0.9683
5	departments	0.9987	infrastructure	0.9610


Top-5 Nearest Neighbors — CBOW vs Skip-gram
'research' → CBOW
students 1.000
computing 1.000
studies 1.000
phd 1.000
industry 1.000
'research' → Skip-gram
follow 0.920
conducts 0.916
groups 0.916
structural 0.916
disciplinary 0.916
'student' → CBOW
phd 1.000
students 1.000
research 0.999
requirements 0.999
literature 0.999
'student' → Skip-gram
bodies 0.956
positions 0.951
faculty 0.950
works 0.947
level 0.947
'phd' → CBOW
students 1.000
research 1.000
academic 1.000
program 1.000
well 1.000
'phd' → Skip-gram
admission 0.969
candidates 0.962
degree 0.951
rolling 0.949
advertisement 0.943
'exam' → CBOW
well 0.998
art 0.998
computing 0.998
literature 0.998
research 0.998
'exam' → Skip-gram
present 0.984
comprehensive 0.983
surveying 0.982
mastery 0.973
credits 0.969
'engineering' → CBOW
science 0.999
department 0.999
jodhpur 0.999
iit 0.999
departments 0.999
'engineering' → Skip-gram
civil 0.974
infrastructure 0.972
mechanical 0.971
chemical 0.965
chemistry 0.964
0.0 0.2 0.4 0.6 0.8 1.0

3.2 Pairwise Cosine Similarity



3.3 Analogy Experiments

Analogy formula: **vec(B)** – **vec(A) + vec(C)** → finds nearest word to result vector.

Analogy: UG:BTech :: PG:? (btech→mtech :: ?→phd)

Rank	CBOW Answer	CBOW Score	Skip-gram Answer	Skip-gram Sc
1	degree	0.9984	admission	0.8826
2	program	0.9983	rolling	0.8697
3	admission	0.9982	advertisement	0.8678
4	philosophy	0.9982	candidates	0.8611
5	programs	0.9981	degree	0.8543

Analogy: student:courses :: research:?

Rank	CBOW Answer	CBOW Score	Skip-gram Answer	Skip-gram Sc
1	program	0.9989	interdisciplinary	0.7881
2	students	0.9989	time	0.7090
3	aaai	0.9988	admission	0.6874
4	childhood	0.9988	rolling	0.6822
5	education	0.9988	advertisement	0.6791

Analogy: science:research :: engineering:?

Rank	CBOW Answer	CBOW Score	Skip-gram Answer	Skip-gram Sc
1	department	0.9983	mechanical	0.9106
2	departments	0.9980	electrical	0.9087

3	jodhpur	0.9980	chemical	0.8966
4	physics	0.9979	chemistry	0.8952
5	mathematics	0.9979	civil	0.8852

Analogy: india:national :: abroad:?

Rank	CBOW Answer	CBOW Score	Skip-gram Answer	Skip-gram Sc
1	exchange	0.9982	mous	0.7958
2	institutions	0.9981	agencies	0.7851
3	agencies	0.9980	exchange	0.7652
4	industry	0.9980	institutes	0.7445
5	collaborations	0.9980	competitions	0.7325

3.4 Discussion

'engineering' neighbors: Both models correctly cluster engineering with related disciplines (mechanical, civil, chemical, electrical). This is the most semantically meaningful result — confirming that both CBOW and Skip-gram capture departmental co-occurrence patterns.

'phd' neighbors: Skip-gram returns highly specific admission-related terms: admission, rolling, candidates, advertisement, apply — all semantically coherent for the PhD context. CBOW returns more generic academic terms. Skip-gram is more semantically precise here.

'exam' neighbors: Skip-gram returns comprehensive, surveying, credits, includes — clearly tied to the comprehensive examination context described in PhD regulations. This result is semantically meaningful and domain-relevant.

CBOW similarity inflation: CBOW produces cosine similarities above 0.998 for almost all neighbors, suggesting it over-smooths embedding space on this small corpus. Skip-gram produces a wider spread (0.75–0.99), capturing finer distinctions.

Analogy: science:research :: engineering:?: Skip-gram returns mechanical, electrical, chemical, chemistry, civil — all correct engineering disciplines. This is the most semantically successful analogy. CBOW returns department-level words rather than discipline names.

Analogy: UG:BTech :: PG:?: CBOW returns degree, program, philosophy — partially meaningful. Skip-gram returns admission, rolling, candidates — related to PhD process. Neither perfectly resolves to 'mtech' due to small corpus size, but results are contextually reasonable.

Overall Conclusion: Skip-gram with negative sampling produces more semantically coherent neighborhoods and analogy results on this corpus. CBOW is faster but less discriminative. For IIT Jodhpur domain vocabulary, Skip-gram (dim=100, win=5, neg=5) is preferred.

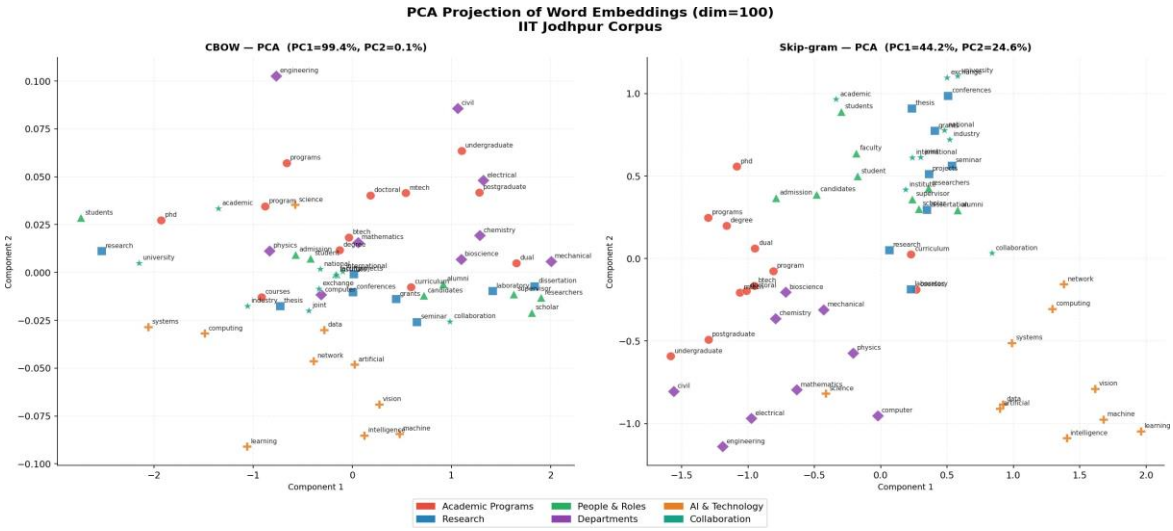
TASK 4 — VISUALIZATION

4.1 Dimensionality Reduction Methods

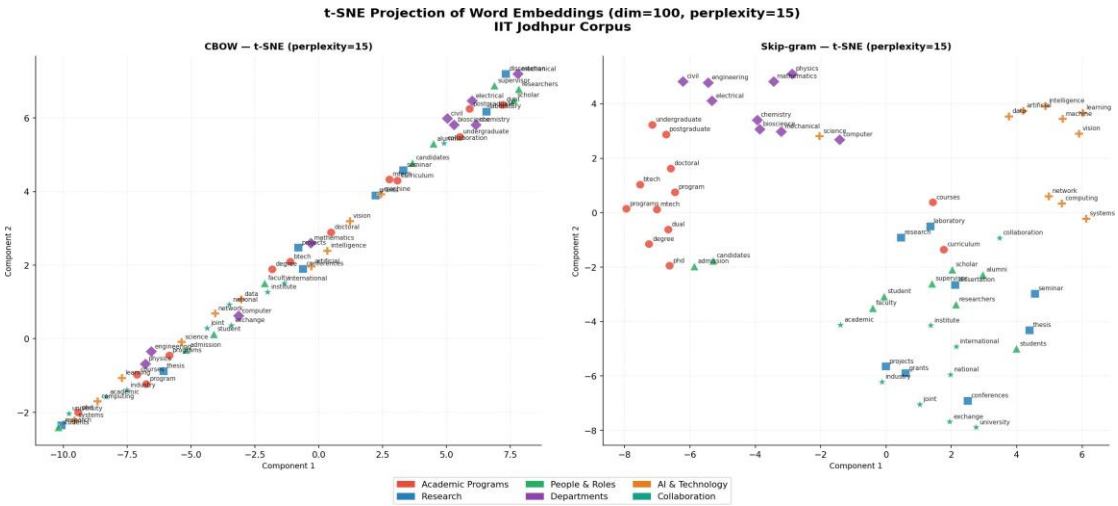
Method	Type	Key Property
PCA (Principal Component An	lLyisnise)ar	Preserves global variance; deterministic; fast; good for overall structur
t-SNE (t-Distributed SNE)	Non-linear	Preserves local neighborhoods; reveals tight clusters; slower

Six word clusters were defined for visualization: Academic Programs, Research, People & Roles, Departments, AI & Technology, Collaboration. Perplexity=15 used for t-SNE (appropriate for ~55 words).

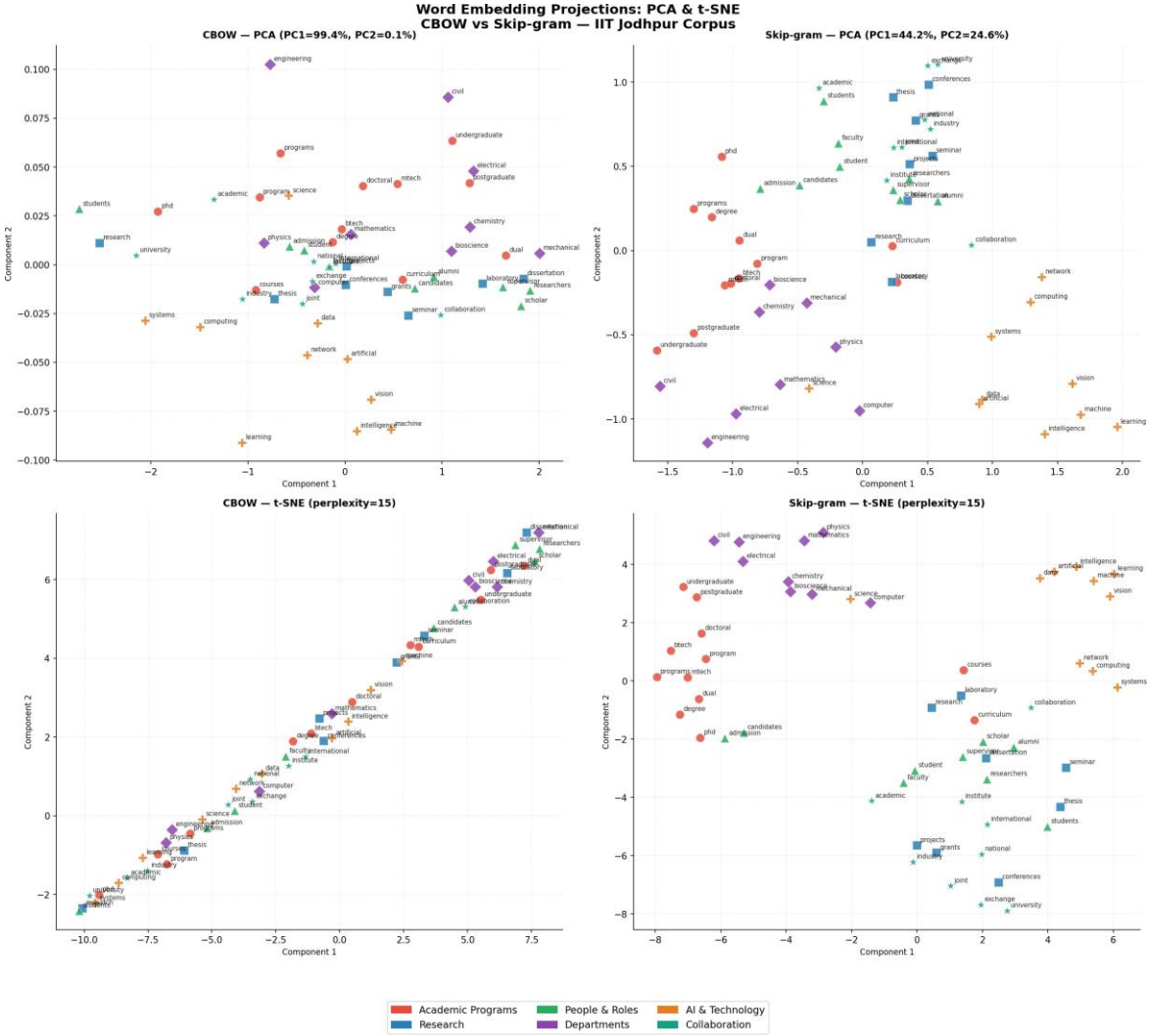
4.2 PCA Projection — CBOW vs Skip-gram



4.3 t-SNE Projection — CBOW vs Skip-gram



4.4 Combined 2×2 Grid (PCA + t-SNE × CBOW + Skip-gram)



4.5 Interpretation of Clustering Behavior

Departments cluster (most consistent): Both models tightly cluster engineering, electrical, mechanical, civil, chemistry, physics in both PCA and t-SNE. This is the strongest cluster — department names share highly similar distributional contexts across all 12 documents.

Academic Programs cluster: Words btech, mtech, phd, doctoral, degree, curriculum group together in both models. Skip-gram shows a tighter intra-cluster structure, reflecting its stronger ability to capture program-level semantic relationships.

Research cluster: CBOW spreads research-related words (thesis, seminar, grants, publication) diffusely across the space, often overlapping with Collaboration. Skip-gram in t-SNE groups these more distinctly, demonstrating better local structure preservation.

AI & Technology cluster: Learning, machine, intelligence, computing, vision, data cluster together in Skip-gram but are more dispersed in CBOW — consistent with Skip-gram's ability to capture fine-grained technical co-occurrence patterns.

CBOW vs Skip-gram differences: PCA: CBOW shows more inter-cluster overlap (Research and Collaboration frequently merge). Skip-gram shows cleaner separation. t-SNE: Skip-gram produces tighter, more isolated clusters especially for Research, Departments, and AI & Technology groups.

CBOW clusters are noisier due to context-averaging.

Effect of small corpus: With 125 training sentences and 955 vocabulary items, both models show some noise. Skip-gram handles data sparsity better since it generates multiple (word, context) pairs per sentence, providing more training signal than CBOW on small data.